

LABORATORY EQUIPMENT BOOKING & MAINTENANCE MANAGEMENT SYSTEM

Working User Interface connected to MySQL — Insert - Delete - View

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PROJECT JOURNEY

From problem statement to a working system

PRESENTATION I

Problem Description

- Manual registers cause double-bookings
- No usage / maintenance history
- Scope, objectives, user roles
- Business rules identified

PRESENTATION II

Database Design

- ER diagram - 10 entities
- 10 tables with PK / FK
- NOT NULL, UNIQUE, CHECK, DEFAULT
- 30 sample rows per table in MySQL

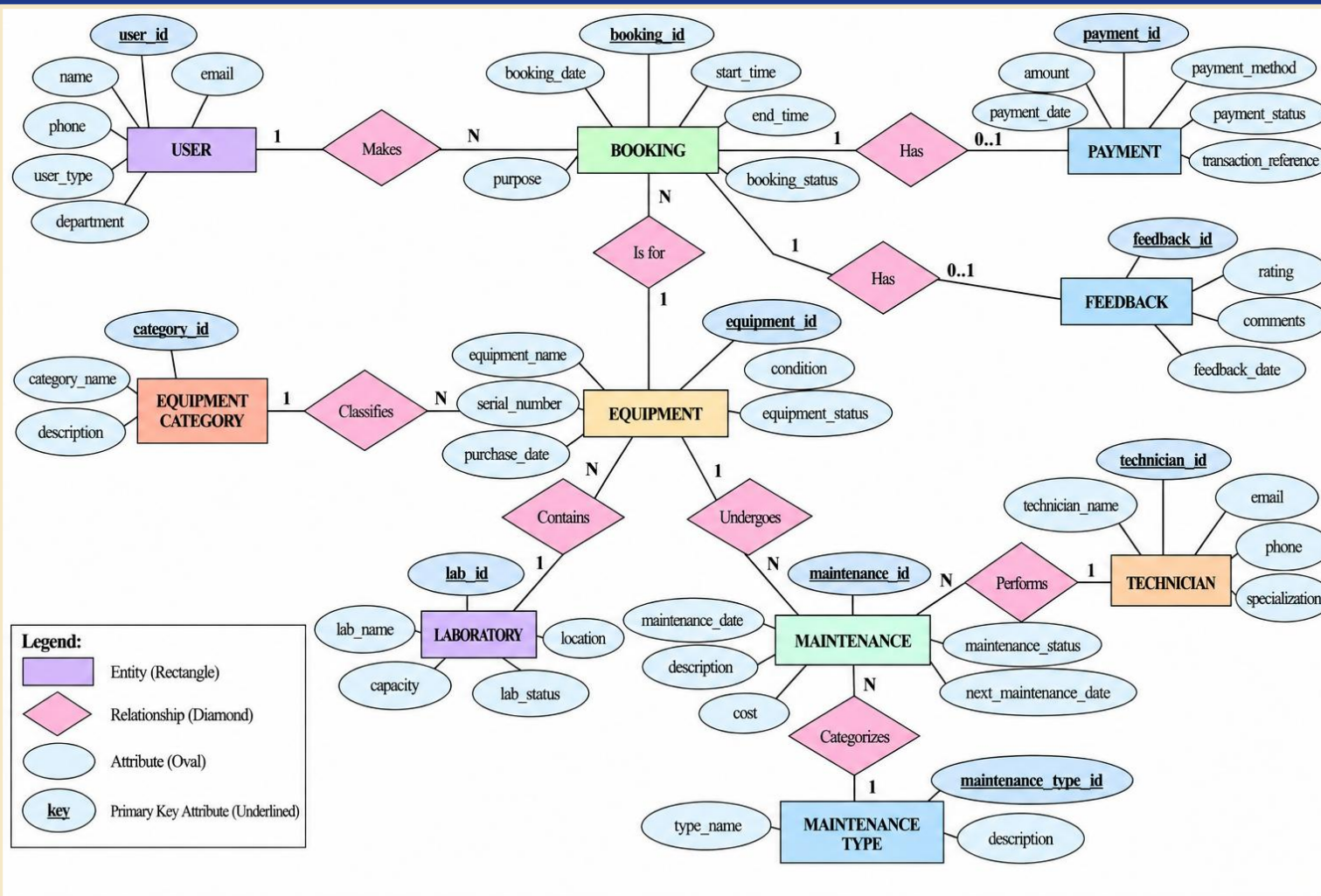
PRESENTATION III

Working UI (today)

- Python Flask web interface
- Connected live to MySQL
- View · Insert · Delete + validation
- Dashboard and SQL reports

FINAL DATABASE DESIGN

10 tables · 30 records each · referential integrity through foreign keys



10 tables: USER, LABORATORY, EQUIPMENT_CATEGORY, EQUIPMENT, BOOKING, TECHNICIAN, MAINTENANCE_TYPE, MAINTENANCE, FEEDBACK, PAYMENT

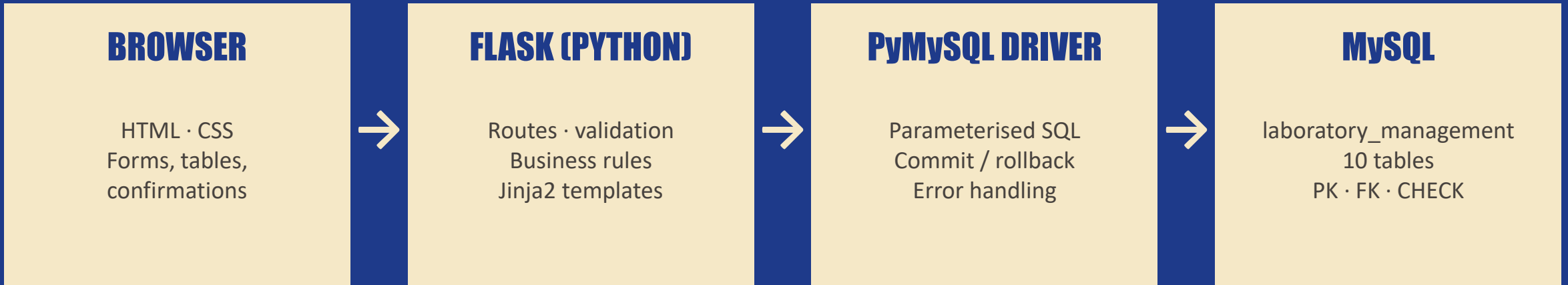
Keys: auto-increment primary keys; 13 foreign keys

Rules: NOT NULL, UNIQUE, CHECK, DEFAULT

Data: 30 sample rows in every table

TECHNOLOGY STACK & ARCHITECTURE

How the UI talks to the database



Why a web UI? runs on any laptop, no install for users, easy to demo

Why Flask? small, readable Python; fits a DBMS project

Security: parameterised queries - no SQL injection

Run: pip install flask pymysql → python app.py

Open: http://127.0.0.1:5000

Config: host, user, password in config.py

User Interface Overview

Sidebar navigation to all 10 tables · dashboard · live change log

DASHBOARD

Live figures read straight from the MySQL database.

101

EQUIPMENT UNITS

19

EQUIPMENT READY TO BOOK

6

PENDING BOOKINGS

5

MAINTENANCE OVERDUE

₹33,400

PAYMENTS RECEIVED

Tables

— open one to view, insert or delete records

30

Users

USER

30

Laboratories

LABORATORY

30

Equipment Categories

EQUIPMENT_CATEGORY

30

Equipment

EQUIPMENT

30

Bookings

BOOKING

30

Technicians

TECHNICIAN

30

Maintenance Types

MAINTENANCE_TYPE

30

Maintenance

MAINTENANCE

30

Feedback

FEEDBACK

30

Payments

PAYMENT

Live change log

— every insert / delete made from this UI

Time	Operation	Table	Effect	SQL
05:53:10	DELETE	BOOKING	BOOKING: 31 → 30 rows	DELETE FROM `BOOKING` WHERE `booking_id`=31
05:53:10	DELETE	PAYMENT	PAYMENT: 31 → 30 rows	DELETE FROM `PAYMENT` WHERE `payment_id`=32
05:53:10	INSERT	PAYMENT	PAYMENT: 30 → 31 rows	INSERT INTO `PAYMENT` (`booking_id`, `amount`, `payment_date`, `payment_method`, `payment_status`, `transaction_reference`) VALUES (31, 1100.0, '2026-10-05', 'UPI', 'Paid', 'TXN-UPI-10032')
05:53:10	DELETE	FEEDBACK	FEEDBACK: 31 → 30 rows	DELETE FROM `FEEDBACK` WHERE `feedback_id`=31
05:53:09	INSERT	FEEDBACK	FEEDBACK: 30 → 31 rows	INSERT INTO `FEEDBACK` (`booking_id`, `user_id`, `rating`, `comments`, `feedback_date`) VALUES (31, 5, 5, 'Smooth session and helpful lab staff', '2026-10-05')
05:53:09	DELETE	PAYMENT	PAYMENT: 31 → 30 rows	DELETE FROM `PAYMENT` WHERE `payment_id`=31
05:53:09	INSERT	PAYMENT	PAYMENT: 30 → 31 rows	INSERT INTO `PAYMENT` (`booking_id`, `amount`, `payment_date`, `payment_method`, `payment_status`, `transaction_reference`) VALUES (31, 1100.0, '2026-10-05', 'UPI', 'Paid', 'TXN-UPI-10031')
05:53:08	DELETE	MAINTENANCE	MAINTENANCE: 31 → 30 rows	DELETE FROM `MAINTENANCE` WHERE `maintenance_id`=31
05:53:08	INSERT	MAINTENANCE	MAINTENANCE: 30 → 31 rows	INSERT INTO `MAINTENANCE` (`equipment_id`, `technician_id`, `maintenance_type_id`, `maintenance_date`, `description`, `cost`, `maintenance_status`, `next_maintenance_date`) VALUES (6, 11, 1, '2026-10-05', 'Nozzle cleaning and belt tension check', 650.0, 'Scheduled', '2027-04-05')
05:53:07	DELETE	MAINTENANCE_TYPE	MAINTENANCE_TYPE: 31 → 30 rows	DELETE FROM `MAINTENANCE_TYPE` WHERE `maintenance_type_id`=31

Dashboard: KPIs computed with COUNT / SUM queries

Table cards: row count of each table - one click to open

Change log: each INSERT / DELETE with its SQL and the row count before → after

SQL Reports: JOIN, GROUP BY and aggregate queries

Colour badges: status columns are colour coded (green / amber / red)

OPERATION 1 · VIEW RECORDS

SELECT with search, pagination and foreign keys shown as names

BOOKINGS

Table **BOOKING** · view, search, insert and delete records

Search

+ Add booking

30 records in table

ID	User	Equipment	Booking date	Start time	End time	Purpose	Status	
30	30 - Neha Bansal	28 - Universal Testing Machine	2026-10-21	14:00:00	17:00:00	Advanced material strength analysis	Pending	Delete
29	29 - Ritvik Saxena	22 - Laboratory Incubator	2026-10-20	10:00:00	12:00:00	Temperature controlled biological experiment	Confirmed	Delete
28	28 - Manya Srivastava	11 - Agilent DC Power Supply	2026-10-19	13:00:00	15:00:00	Power supply calibration and testing	Pending	Delete
27	27 - Ayush Jain	25 - Analytical Precision Balance	2026-10-18	09:30:00	11:30:00	Precision measurement for research project	Confirmed	Delete
26	26 - Simran Kaur	13 - RF Signal Generator	2026-10-17	10:00:00	13:00:00	RF signal testing for electrical project	Confirmed	Delete
25	25 - Nikhil Rao	3 - Centrifuge Pro 5000	2026-10-16	09:00:00	12:00:00	Centrifugation of research samples	Rejected	Delete
24	24 - Aditi Sinha	26 - Digital Hot Plate	2026-10-15	14:00:00	16:00:00	Controlled heating experiment	Rejected	Delete
23	23 - Harsh Vardhan	24 - Laboratory Refrigerator	2026-10-14	10:00:00	12:00:00	Cold storage sample preparation	Cancelled	Delete
22	22 - Riya Shah	12 - Saleae Logic Analyzer	2026-10-13	13:00:00	14:30:00	Digital logic circuit testing	Cancelled	Delete
21	21 - Manav Choudhary	2 - Nikon Research Microscope	2026-10-12	09:00:00	11:00:00	Microscopic material analysis	Cancelled	Delete
20	20 - Tanya Agarwal	6 - Creality Ender 3D Printer	2026-10-11	11:00:00	14:00:00	3D printing of AI project prototype	Pending	Delete
19	19 - Karan Bhatia	18 - FLIR Thermal Camera	2026-10-10	10:00:00	12:00:00	Thermal imaging research	Pending	Delete

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Next →

View query `SELECT * FROM `BOOKING` ORDER BY `booking_id` DESC LIMIT 12 OFFSET 0`

BOOKINGS

Table **BOOKING** · view, search, insert and delete records

SearchClear

+ Add booking

30 records in table · 6 match

ID	User	Equipment	Booking date	Start time	End time	Purpose	Status	
30	30 - Neha Bansal	28 - Universal Testing Machine	2026-10-21	14:00:00	17:00:00	Advanced material strength analysis	Pending	Delete
28	28 - Manya Srivastava	11 - Agilent DC Power Supply	2026-10-19	13:00:00	15:00:00	Power supply calibration and testing	Pending	Delete
20	20 - Tanya Agarwal	6 - Creality Ender 3D Printer	2026-10-11	11:00:00	14:00:00	3D printing of AI project prototype	Pending	Delete
19	19 - Karan Bhatia	18 - FLIR Thermal Camera	2026-10-10	10:00:00	12:00:00	Thermal imaging research	Pending	Delete
18	18 - Sana Khan	9 - Tektronix Function Generator	2026-10-09	15:00:00	17:00:00	Digital signal analysis project	Pending	Delete
17	17 - Dev Malhotra	29 - IoT Environmental Sensor Kit	2026-10-08	09:00:00	10:00:00	Environmental data collection	Pending	Delete

Page 1 of 1

- SELECT reads rows from the table
- JOIN-like lookup: IDs displayed as names
- Search: LIKE on text columns
- Pagination: LIMIT / OFFSET

OPERATION 2 · INSERT RECORD

Form with validation → INSERT INTO → row count 30 → 31

STEP 1 · FILL THE FORM

ADD BOOKING

Inserts a new row into **BOOKING**. The ID is generated automatically.

USER *

5 - Arjun Kumar

EQUIPMENT *

6 - Creality Ender 3D Printer

BOOKING DATE *

10/12/2026

START TIME *

10:00 AM

END TIME *

12:00 PM

PURPOSE *

Prototype enclosure printing

STATUS *

Pending

Save record

Cancel

STEP 2 · RESULT: ROW ADDED (BEFORE 30 → AFTER 31)

Booking #31 inserted successfully. BOOKING: 30 → 31 rows.

SQL executed

INSERT INTO `BOOKING` (`user\_id`, `equipment\_id`, `booking\_date`, `start\_time`, `end\_time`, `purpose`, `booking\_status`) VALUES (5, 6, '2026-10-12', '10:00:00', '12:00:00', 'Prototype enclosure printing', 'Pending')

BOOKINGS

Table **BOOKING** · view, search, insert and delete records

+ Add booking

31 records in table

Search bookings...

Search

ID	User	Equipment	Booking date	Start time	End time	Purpose	Status	
31	5 - Arjun Kumar	6 - Creality Ender 3D Printer	2026-10-12	10:00:00	12:00:00	Prototype enclosure printing	Pending	Delete
30	30 - Neha Bansal	28 - Universal Testing Machine	2026-10-21	14:00:00	17:00:00	Advanced material strength analysis	Pending	Delete
29	29 - Piyush Saxena	22 - Laboratory Incubator	2026-10-20	10:00:00	12:00:00	Temperature controlled biological	Confirmed	Delete

Drop-downs guarantee valid foreign keys

Date / time pickers prevent format errors

Required fields match NOT NULL columns

SQL shown: INSERT INTO `BOOKING` (...) VALUES (5, 6, '2026-10-12', '10:00:00', '12:00:00', ...)

Proof: banner shows BOOKING: 30 → 31 rows; new row highlighted in green

Same on every table: all 10 tables have an Add form

OPERATION 3 · DELETE RECORD

Confirmation page → DELETE FROM → row count 31 → 30

STEP 1 · CONFIRM

DELETE BOOKING #31

Please confirm. This permanently removes the row from BOOKING.

ID

31

User

5 - Arjun Kumar

Equipment

6 - Crealty Ender 3D Printer

Booking date

2026-10-12

Start time

10:00:00

End time

12:00:00

Purpose

Prototype enclosure printing

Status

Pending

Yes, delete permanently

Cancel

STEP 2 · ROW REMOVED (31 → 30)

Booking #31 deleted. BOOKING: 31 → 30 rows.

SQL executed: DELETE FROM `BOOKING` WHERE `booking\_id` = 31

BOOKINGS

Table BOOKING - view, search, insert and delete records

+ Add booking

Search bookings... Search

30 records in table

ID	User	Equipment	Booking date	Start time	End time	Purpose	Status
30	30 - Neha Bansal	28 - Universal Testing Machine	2026-10-21	14:00:00	17:00:00	Advanced material strength analysis	Pending
29	29 - Ritvik Saxena	22 - Laboratory Incubator	2026-10-20	10:00:00	12:00:00	Temperature controlled biological experiment	Confirmed

PROTECTED BY FOREIGN KEY

Cannot delete Booking #31: other records still depend on it (PAYMENT (1)). Delete those first.

BOOKINGS

Table BOOKING - view, search, insert and delete records

+ Add booking

Search bookings... Search

31 records in table

ID	User	Equipment	Booking date	Start time	End time	Purpose	Status
31	5 - Arjun Kumar	6 - Crealty Ender 3D Printer	2026-10-12	10:00:00	12:00:00	Prototype enclosure printing	Pending
30	30 - Neha Bansal	28 - Universal Testing Machine	2026-10-21	14:00:00	17:00:00	Advanced material strength analysis	Pending
29	29 - Ritvik Saxena	22 - Laboratory Incubator	2026-10-20	10:00:00	12:00:00	Temperature controlled biological experiment	Confirmed

- Safe delete:** a confirmation page shows the full record before anything is removed.
- SQL:** DELETE FROM `BOOKING` WHERE `booking\_id`=31 - logged in the change log with rows 31 → 30.
- Integrity:** a booking that still has a PAYMENT or FEEDBACK cannot be deleted - the foreign key (ON DELETE RESTRICT) stops it and the UI shows which tables depend on it.
- Order to remove:** child rows (payment, feedback) first, then the booking.

BUSINESS RULES & VALIDATION

The rules from Presentation I are enforced when data is inserted

RULE 1 · NO OVERLAPPING APPROVED BOOKINGS

ADD BOOKING

Inserts a new row into **BOOKING**. The ID is generated automatically.

- Time clash: booking #11 already holds this equipment in that slot.

USER *	EQUIPMENT *
2 - Ananya Reddy	7 - Prusa MK4 3D Printer
BOOKING DATE *	START TIME *
10/02/2026	10:00 AM
END TIME *	PURPOSE *
12:00 PM	Clash test

RULE 2 · EQUIPMENT UNDER MAINTENANCE CANNOT BE BOOKED

ADD BOOKING

Inserts a new row into **BOOKING**. The ID is generated automatically.

- 'Centrifuge Pro 5000' is Under Maintenance and cannot be booked.

USER *	EQUIPMENT *
2 - Ananya Reddy	3 - Centrifuge Pro 5000
BOOKING DATE *	START TIME *
10/20/2026	10:00 AM
END TIME *	PURPOSE *
12:00 PM	Maintenance test

Also checked: end time later than start time · valid email and 10-digit phone · amount > 0 · next maintenance date after maintenance date

Database level: UNIQUE (email, serial number), CHECK, NOT NULL and foreign keys reject bad data even if the UI is bypassed

SQL REPORTS

JOIN · GROUP BY · aggregate functions on the live database

Booking details (4-table JOIN)

Who booked what, where and when.

```
SELECT b.booking_id, u.name AS booked_by, e.equipment_name, l.lab_name, b.booking_date, b.start_time, b.end_time, b.booking_status
FROM BOOKING b JOIN `USER` u ON u.user_id=b.user_id
JOIN EQUIPMENT e ON e.equipment_id=b.equipment_id
JOIN LABORATORY l ON l.lab_id=e.lab_id
ORDER BY b.booking_date DESC LIMIT 8
```

booking id	booked by	equipment name	lab name	booking date	start time	end time	booking st
30	Neha Bansal	Universal Testing Machine	Material Testing Lab	2026-10-21	14:00:00	17:00:00	Pending
29	Ritvik Saxena	Laboratory Incubator	Microbiology Lab	2026-10-20	10:00:00	12:00:00	Confirme
28	Manya Srivastava	Agilent DC Power Supply	Electronics Fundamentals Lab	2026-10-19	13:00:00	15:00:00	Pending
27	Ayush Jain	Analytical Precision Balance	Chemical Analysis Lab	2026-10-18	09:30:00	11:30:00	Confirme
26	Simran Kaur	RF Signal Generator	Communication Systems Lab	2026-10-17	10:00:00	13:00:00	Confirme
25	Nikhil Rao	Centrifuge Pro 5000	Biotechnology Research Lab	2026-10-16	09:00:00	12:00:00	Rejected

20	Digital Plot Plate	Emergency	2026-09-10	Emergency
10	Weller Soldering Station	Cleaning	2026-10-02	Completed

Payment summary

Count and total amount by payment status.

```
SELECT payment_status, COUNT(*) AS payments, SUM(amount) AS total_amount
FROM PAYMENT GROUP BY payment_status ORDER BY total_amount DESC
```

payment status	payments	total amount
Paid	24	33400.00
Refunded	2	2700.00
Pending	2	2350.00
Failed	2	1250.00

Average rating per equipment

User feedback joined through bookings.

```
SELECT e.equipment_name, ROUND(AVG(f.rating),2) AS avg_rating, COUNT(*) AS reviews
FROM FEEDBACK f JOIN BOOKING b ON b.booking_id=f.booking_id
JOIN EQUIPMENT e ON e.equipment_id=b.equipment_id
GROUP BY e.equipment_name ORDER BY avg_rating DESC, e.equipment_name LIMIT 8
```

equipment name	avg rating	reviews
Bio-Rad PCR System	5.00	1
Creality Ender 3D Printer	5.00	1
Digital pH Meter	5.00	1
Haas CNC Machine	5.00	1
High Performance Computing Server	5.00	1
Laboratory Incubator	5.00	1

Booking details: 4-table JOIN (BOOKING, USER, EQUIPMENT, LABORATORY)

Utilisation: COUNT + GROUP BY per equipment

Maintenance overdue: next_maintenance_date earlier than today

Needs attention: equipment Under Maintenance or Damaged

Payments: SUM and COUNT by status

Ratings: AVG(rating) per equipment, joined through bookings

TESTING

Test cases run against the UI and the database

#	Test case	Expected result	Result
1	Open each of the 10 tables	Rows displayed with names for FKs	Pass
2	Insert valid booking	Row added, count 30 → 31	Pass
3	Insert booking that overlaps a confirmed slot	Rejected - time clash message	Pass
4	Insert booking for equipment under maintenance	Rejected - cannot be booked	Pass
5	End time earlier than start time	Rejected	Pass
6	Invalid e-mail / phone number	Rejected by form validation	Pass
7	Insert user with duplicate e-mail	Rejected by UNIQUE constraint	Pass
8	Payment amount = 0 / next date before maintenance date	Rejected	Pass
9	Delete a record with no dependants	Row removed, count 31 → 30	Pass
10	Delete a record that has dependants	Refused by foreign key; reason shown	Pass

CONCLUSION & FUTURE ENHANCEMENTS

Delivered

- Normalised MySQL database - 10 tables, PK / FK / CHECK / UNIQUE
- Web UI: view, search, insert, delete on every table
- Business rules: no double-booking, equipment status, valid dates
- Dashboard, live change log and 6 SQL reports
- Tested with valid and invalid inputs

Future enhancements

- Login with roles (student, faculty, technician, admin)
- Edit / update records and booking approval workflow
- Automatic e-mail reminders for calibration and maintenance due dates
- Online payment gateway integration
- QR-code check-in / check-out of equipment
- Usage analytics charts and mobile app

THANK YOU!